# Automated Scoring of Short Answer Questions with Large Language Models: Impacts of Model, Item, and Rubric Design

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github.com/Khan/llm-saq-scoring

# Constructed Response Items

Include Essays and Short Answer Questions (SAQs).

Can reduce the impact of guessing [1].

Can assess complex skills [1].

Need for humans to grade responses on a rubric.

[1] Livingston S.A.: Constructed-Response Test Questions: Why We Use Them; How We Score Them. (2009).

# Operational Definitions

**Short Answer Questions**:
- responses up to 3 sentences
- 2-3 scoring levels

**Rubric**:
- scoring criteria
- graded response examples

# Automated Scoring (AS)

Saves time and resources for scoring CR items

A model is created to score CR items, and humans only score a subset of items, for training & QA.

Methods have increased in sophistication and accuracy over time:

- NLP techniques w/ feature engineering [2]
- Deep Learning techniques [3]
- **Finetuned** LLMs [8-9]

# **Off-The-Shelf** LLMs for AS

Pre-trained model, ready-to-use without additional customization (e.g., GPT-4)

+ Scoring using **prompt engineering**
+ Promising performance so far [10-12]
+ Potential to score CR items at scale by further saving time and resources (though we still need human raters)

- LLM “Black Box” hinders explainability [16]
- Potential for bias in scoring [14]
- Cost, speed


[2] Landauer T.K., Laham D., Rehder B., Schreiner M.E.: How well can passage meaning be derived without using word order? A comparison of Latent Semantic Analysis and humans. (1997)
[3] Taghipour K., Ng H.T.: A neural approach to automated essay scoring. (2016)
[8] Morris W., Holmes L., Choi J.S., Crossley S.: Automated scoring of constructed response items in math assessment using large language models. (2024).
[9] Kwako, A., & Ormerod, C. Improving transformer-based automated scoring of K-12 science items. (2024).
[10] Henkel, O., Hills, L., Boxer, A., Roberts, B., Levonian, Z.: Can large language models make the grade? An empirical study evaluating LLMs' ability to mark short-answer questions in K-12 education. (2024)
[11] Pack A., Barrett A., Escalante J.: Large language models and automated essay scoring of English language learner writing: Insights into validity and reliability. (2024).
[12] Lee G.G., Latif E., Wu X., Liu N., Zhai X.. Applying large language models and chain-of-thought for automatic scoring. (2024)
[14] Johnson M., Zhang M.: Examining the responsible use of zero-shot AI approaches to scoring essays. (2024)
[16] Casabianca, J.M., McCaffrey, D.F., Johnson, M.S., Alper, N., Zubenko, V.: Validity Arguments For Constructed Response Scoring Using Generative Artificial Intelligence Applications (2024)

## Finetuning LLMs for AS

***Morris et al (2024)***
G4 & G8 math items, 2-3 score levels
DeBERTa approached human-level agreement

***Kwako and Ormerod (2024)***
G4-10 science Items, 2-3 score levels
DeBERTa did best, approaching human-level agreement for some items.

## Prompt Engineering for AS

***Lee et al (2024)***
Middle school science items, 2-3 score levels
GPT-4 had good accuracy for most dichotomous items

***Henkel et al (2024)***
~Grade 3-11 science & history items, 2 score levels
GPT-4 did well, approached human-level agreement

## Gaps in the Literature

Research often uses existing items & data

Studies often focus on one or a few models

Less research on how rubrics influence scoring performance

Less research on how item characteristics influence LLM scoring performance

# The Current Study

Purpose is to evaluate the performance of different LLMs for scoring SAQs under different conditions.

(RQ1) How much rubric information is needed in the prompt to produce reliable LLM ratings?

(RQ2) How well do various LLMs score SAQs?

(RQ3) What are the features of SAQs with the best and worst LLM performance?



# Method

# Items

SMEs created **10 HS English Language Arts (ELA)** and **10 HS Algebra 1 (Math)** SAQs w/ the purpose of LLM scoring.

Items were dichotomous (2 score levels: Correct or Incorrect), and contained:

1. A stem,
2. Scoring criteria (for correct),
3. Examples of correct answers, and
4. Examples of incorrect answers

*Rubric Elements*

We classified all items along four dimensions:

**1. Response Length**
- **Shorter:** Less than 1 sentence
- **Longer:** 1 sentence or more

**2. Cognitive Demand** (Bloom's Taxonomy)
- **Low:** Remember, Understand, Apply
- **High:** Analyze, Evaluate, Create

**3. Scoring Subjectivity**
- **Low:** Straightforward matching against clear criteria
- **High:** Some judgment required within defined parameters

**4. Answer Scope**
- **Narrow:** Very few correct answers
- **Moderate:** Several acceptable correct answers
- **Broad:** Many possible correct answers

# Example: Item 20 (Math)

Response Length: **Longer** // Cognitive Complexity: **High** // Scoring Subjectivity: **Low** // Answer Scope: **Moderate**

| | |
|---|---|
| **Stem** | Taylor is making friendship bracelets that she plans to sell on an Etsy shop next year. In addition to her current stock of bracelets, she wants to consistently make bracelets to add to her shop each month. The expression 50m + 230 represents the total number bracelets available to sell after m months.<br><br>Explain what the values in the expression mean in the real-world scenario. |
| **Scoring Criteria** | - Student recognizes that the coefficient of m, 50, represents a rate. In the context of the problem, the rate is the number of bracelets added to the shop each month.<br>- Student recognizes that the constant term 230 represents a starting point before any time elapses. In the context of the problem, the starting point is the current stock of bracelets. |
| **Correct Answer Example(s)** | “The coefficient of m, 50, represents the number of bracelets that Taylor makes monthly. The term 230 represents the current stock of bracelets before adding to the stock monthly.” |
| **Incorrect Answer Example(s)** | "Taylor makes bracelets for 50 months and ends up with 230 bracelets in total." |

| Domain | Item | Description | Response Length | Cognitive Demand | Answer Scope | Scoring Subjectivity |
|---|---|---|---|---|---|---|
| ELA | 1 | Identify meaning of a word in a passage | Shorter | Low | Narrow | Low |
| | 2 | Interpret tone of passage | Shorter | High | Narrow | Low |
| | 3 | Identify 3 words from passage with negative connotation | Shorter | High | Moderate | Low |
| | 4 | Identify author's purpose for writing a passage | Longer | High | Moderate | Low |
| | 5 | Describe how an idea is developed in passage | Longer | High | Moderate | High |
| | 6 | Identify meaning of an idiom in passage context | Shorter | Low | Narrow | Low |
| | 7 | Summarize the central idea of a passage | Longer | Low | Moderate | High |
| | 8 | Identify an author's primary argument using evidence | Longer | High | Broad | High |
| | 9 | Summarize a text | Longer | Low | Broad | High |
| | 10 | Determine if a text supports a theme using evidence | Longer | High | Broad | High |
| Math | 11 | Identify 5 monomial factors of a polynomial with two variables | Shorter | Low | Broad | Low |
| | 12 | Write an equation with infinite solutions | Shorter | Low | Broad | Low |
| | 13 | Write a polynomial in proper form | Shorter | Low | Broad | Low |
| | 14 | Identify two ways to increase value of expression by changing two variables | Shorter | Low | Narrow | Low |
| | 15 | Identify approach for solving a system of simple equations | Longer | Low | Moderate | Low |
| | 16 | Explain why a given function is a function | Longer | Low | Moderate | High |
| | 17 | Give triangle measurements that falsify a statement about angle bisectors | Longer | High | Broad | Low |
| | 18 | Describe real-world context and key points for an inverted parabola equation | Longer | High | Broad | High |
| | 19 | Interpret the meaning of c(0) for a given function in a real-world scenario | Longer | High | Moderate | Low |
| | 20 | Explain values of an expression in a real-world scenario | Longer | High | Moderate | Low |

# Responses

Created ~**60 incorrect responses** to each item (~1,200 total), similar to those expected of the target population.

Reduced to **20 distinct incorrect responses** for each item.

Used OpenAI o1 to generate **20 correct responses**, matched to the incorrect responses in terms of tone and style

Final dataset included **40 responses** to each item (**~20 incorrect**, **~20 correct**), 800 responses total

[1] "50 is the number of bracelets she makes each month, and 230 is the number of bracelets she already has."
[2] "She makes 50 bracelets a month on average."
[3] "m is months, so she works for 50 months, and 230 is number of bracelets, so after 50 months she has 230 bracelets to sell"
[4] "It means she has 230 bracelets at the beginning, and she makes 50 bracelets each month."
[5] "She sells 50 bracelets to make $230 so 230/50 is the price"
[6] "50 is bracelets per month, so she makes 50 bracelets every month, and 230 is number of bracelets she has before, so before she starts adding more, she has 230 bracelets"
[7] "It means Taylor adds 50 bracelets to her shop each month, and she starts with 230 bracelets."
[8] "50m + 230 is the total number of bracelets"
[9] "50 represents the number of bracelets Taylor makes every month. 230 represents the bracelets she already has before she begins adding more."
[10] "The value is 50m + 230 so 50 is the rate of bracelets added each month and 230 is the starting amount"
...

# Scoring

**Human Scoring**:

- Every response was labeled by 3 SMEs
- Scored according to **Full Rubric** (Stem, Scoring Criteria, and Examples)
- Majority vote was used for the **Ground Truth**

**LLM Scoring**:

We prompted 11 LLMs to score each response using 3 different rubrics:

- **Empty**: Only the Stem was provided
- **Criteria Only:** The Stem and Scoring Criteria were provided (akin to “Zero-Shot”)
- **Full:** The Stem, Scoring Criteria, and Examples were provided (akin to “Few-Shot”)

Each LLM gave 3 scores for every response for each rubric

- One API call, temperature = 0.30 (except for OpenAI o1, which is fixed at 1)
- Majority vote was used as the **Predicted Score**
- 3 rubrics x 11 models = 33 sets of scores

## LLMs

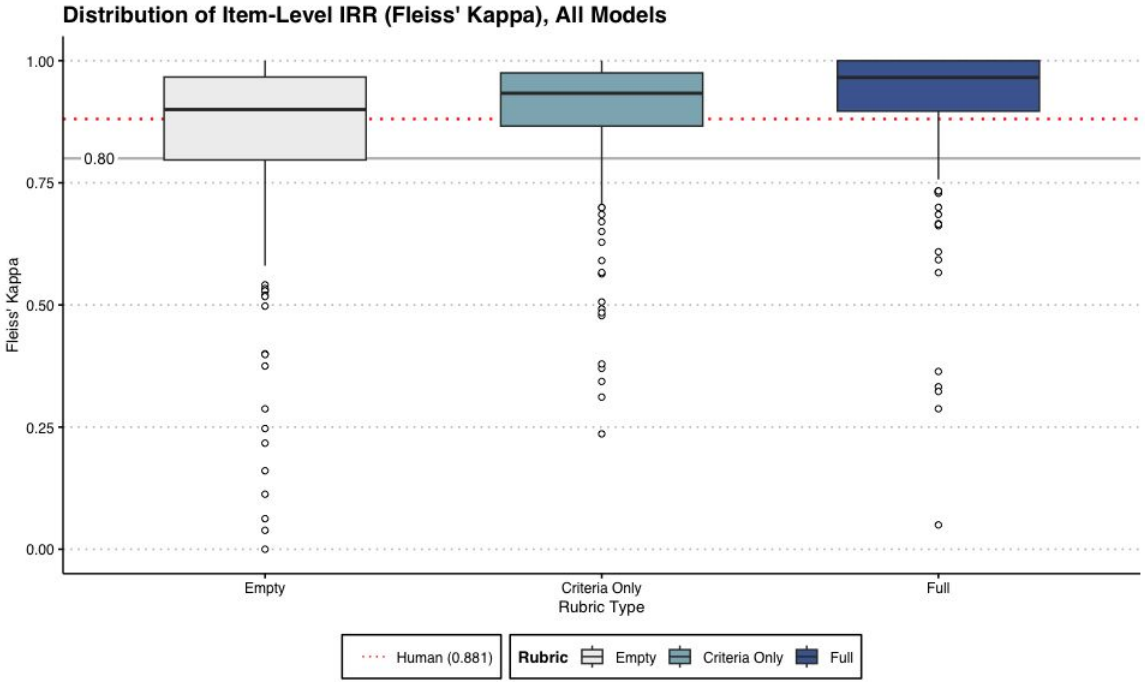
**Small**
Claude 3.5 Haiku
Gemini 1.5 Flash
Gemini 1.5 Flash 8b
GPT-4o mini
Llama 3.1 70b
Llama 3.1 8b

**Large**
Claude 3.5 Sonnet
Gemini 1.5 Pro
GPT-4o
Llama 3.1 405b

**Frontier**
OpenAI o1

# Results

# Interrater Reliability by Rubric




RQ1: How much rubric information is needed in the prompt to produce reliable LLM ratings?

[15] Landis J.R., Koch G.G.. The measurement of observer agreement for categorical data. (1977)

**Notes**

- Interrater Reliability (IRR) w/ Fleiss' Kappa, 3 ratings
- Kappa > 0.80 represents Almost Perfect Agreement [15]
- Human IRR (Full Rubric) = 0.881
- Each bar plots 11 LLMs x 20 items (n = 220)

**Results**

**ANOVA**: $F(2, 657) = 9.98$, $p < .001$; $\eta^2 = 0.03$

**Contrasts:**

***Criteria Only > Empty**

$M_{diff} = -0.05$, $SE = 0.015$, $t(657) = -3.37$, $p < .001$

***Full > Empty**

$M_{diff} = -0.063$, $SE = 0.015$, $t(657) = -4.23$, $p < .001$

Full v. Criteria Only

$M_{diff} = -0.013$, $SE = 0.015$, $t(657) = -0.86$, $p = .391$

# Overall Interrater Reliability by Rubric and Model

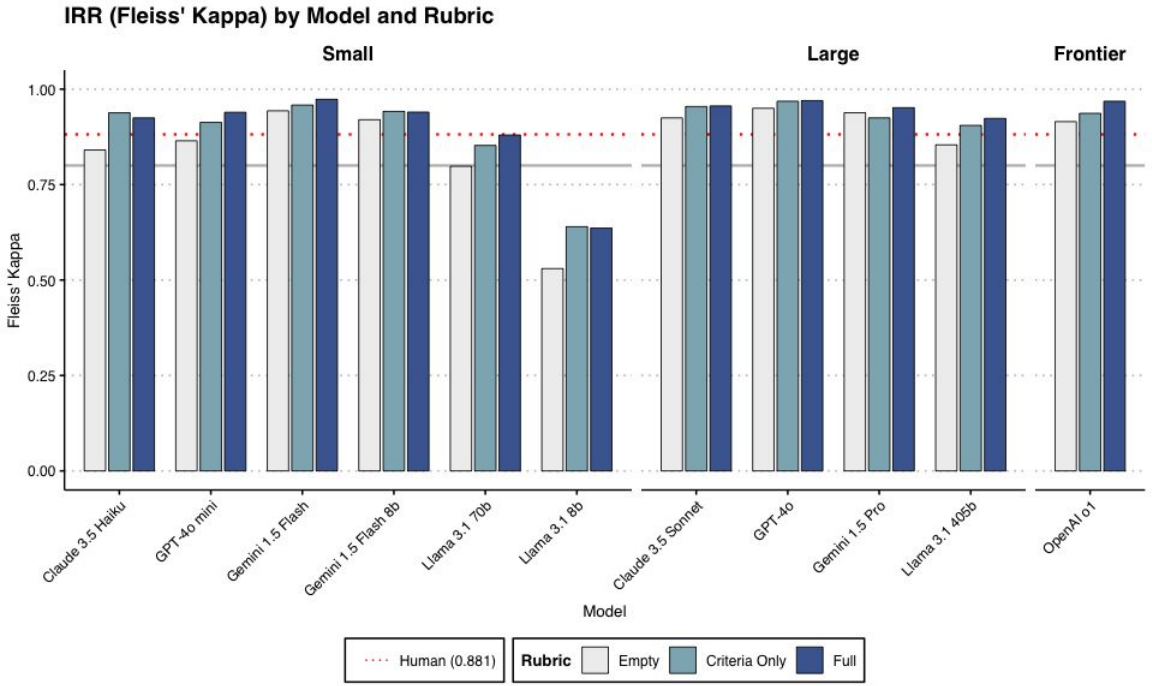
| | | IRR (Fleiss' Kappa) | | |
|---|---|---|---|---|
| **Size** | **LM** | **Empty** | **Crit** | **Full** |
| Small | Claude-3-5 Haiku | .841 | .938 | .925 |
| | Gemini-1.5 Flash | .943 | .958 | .973 |
| | Gemini-1.5 Flash 8b | .920 | .942 | .940 |
| | GPT 4o-mini | .865 | .913 | .939 |
| | Llama 3.1 70b | .798 | .853 | .880 |
| | Llama 3.1 8b | .530 | .640 | .636 |
| Large | Claude 3-5 Sonnet | .925 | .954 | .956 |
| | Gemini 1.5 Pro | .938 | .925 | .951 |
| | GPT-4o | .950 | .968 | .970 |
| | Llama 3.1 405b | .854 | .905 | .923 |
| Frontier | Open AI o1 | .915 | .937 | .968 |

RQ1: How much rubric information is needed in the prompt to produce reliable LLM ratings?

**Notes**

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# Overall LLM Interrater Reliability by Rubric and Model




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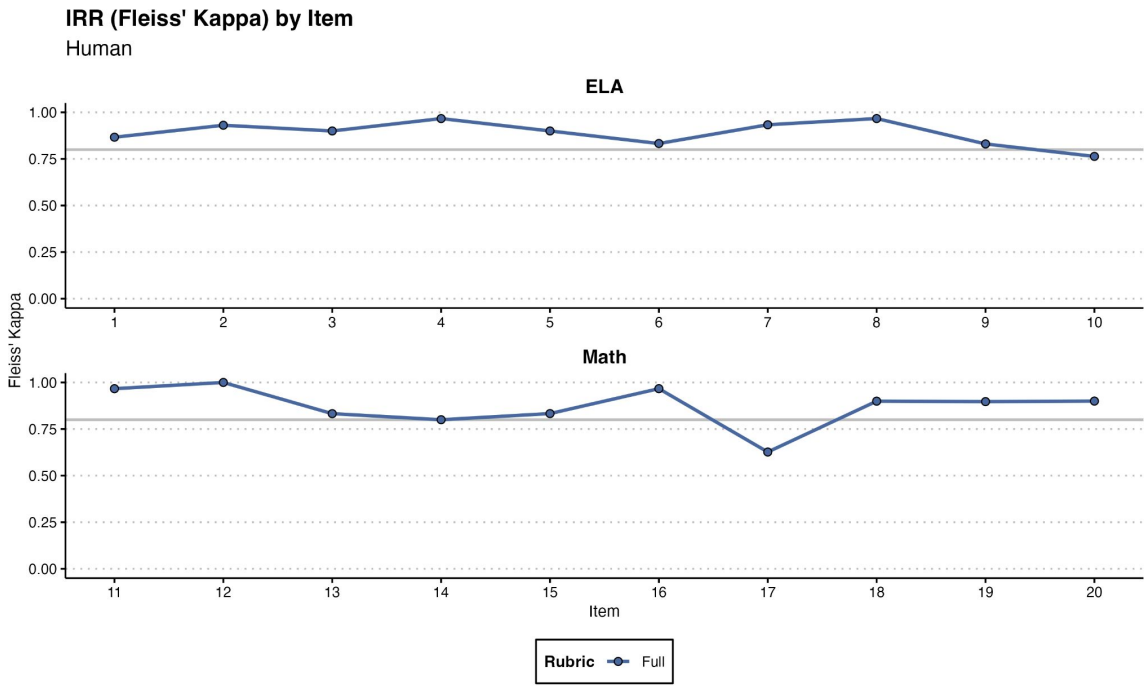
**Notes**

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**Observations**

- Most LLMs had IRR > Human for at least 1 rubric type
- Some LLMs had IRR > Human with Empty rubrics (no scoring criteria or examples)
- More robust rubrics >> better LLM interrater reliability

# Item-Level LLM Interrater Reliability by Rubric




RQ1: How much rubric information is needed in the prompt to produce reliable LLM ratings?

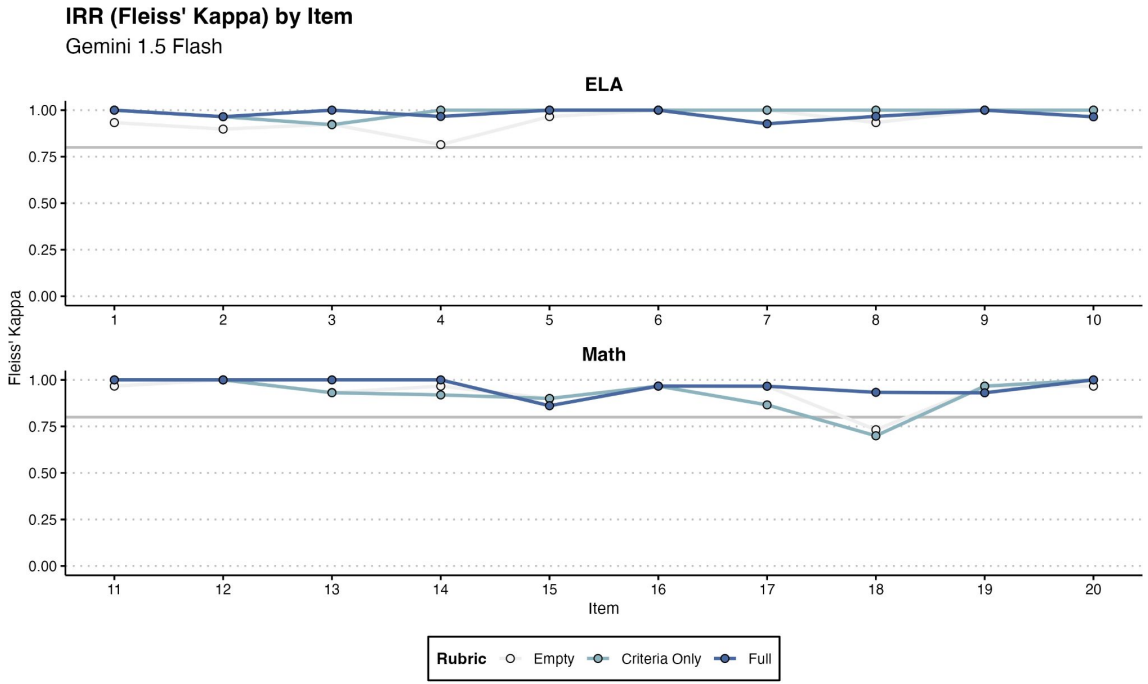
**Notes**

- LLM Temperature = 0.3
- Human IRR (Full Rubric) = 0.881
- Kappa > 0.80 represents Almost Perfect Agreement [15]
- Each bar plots 11 LLMs

**Observations**

- IRR varied by item
- Some items had almost perfect agreement across LLM and rubric
- Some items had generally low IRR

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RQ1: How much rubric information is needed in the prompt to produce reliable LLM ratings?

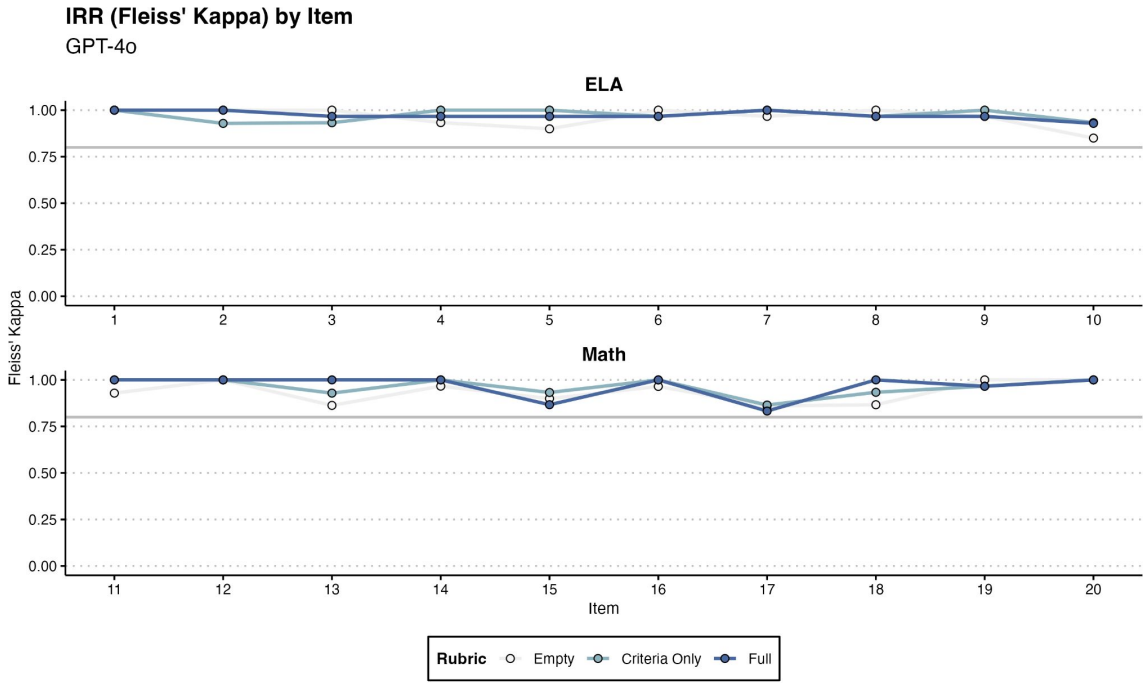
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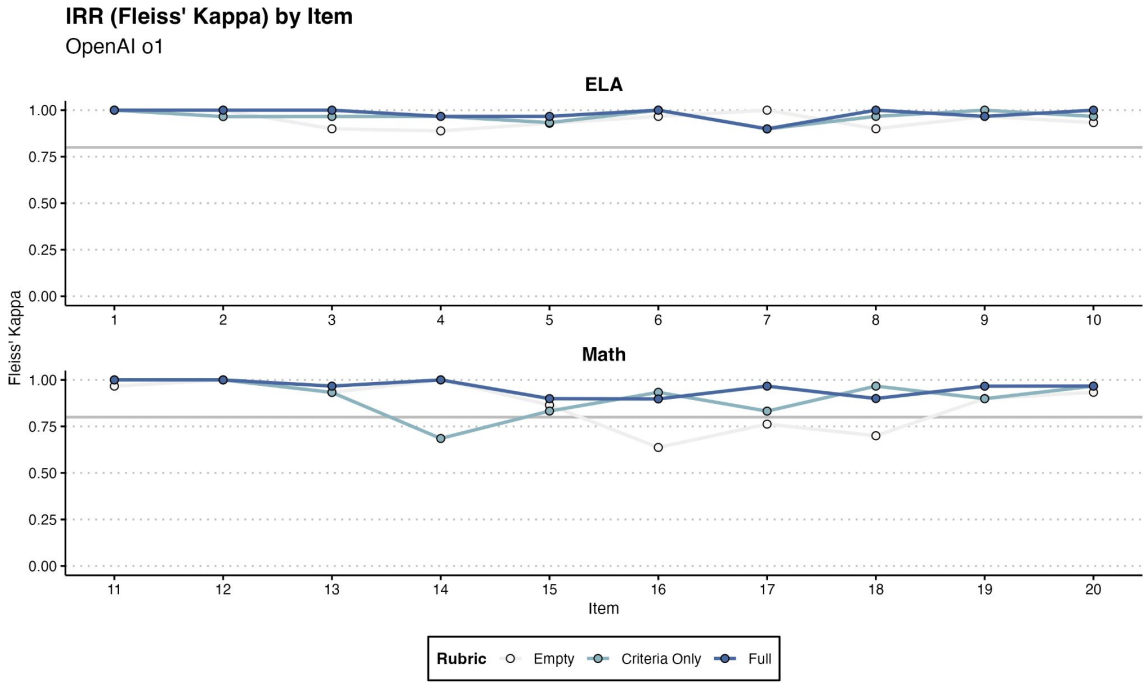
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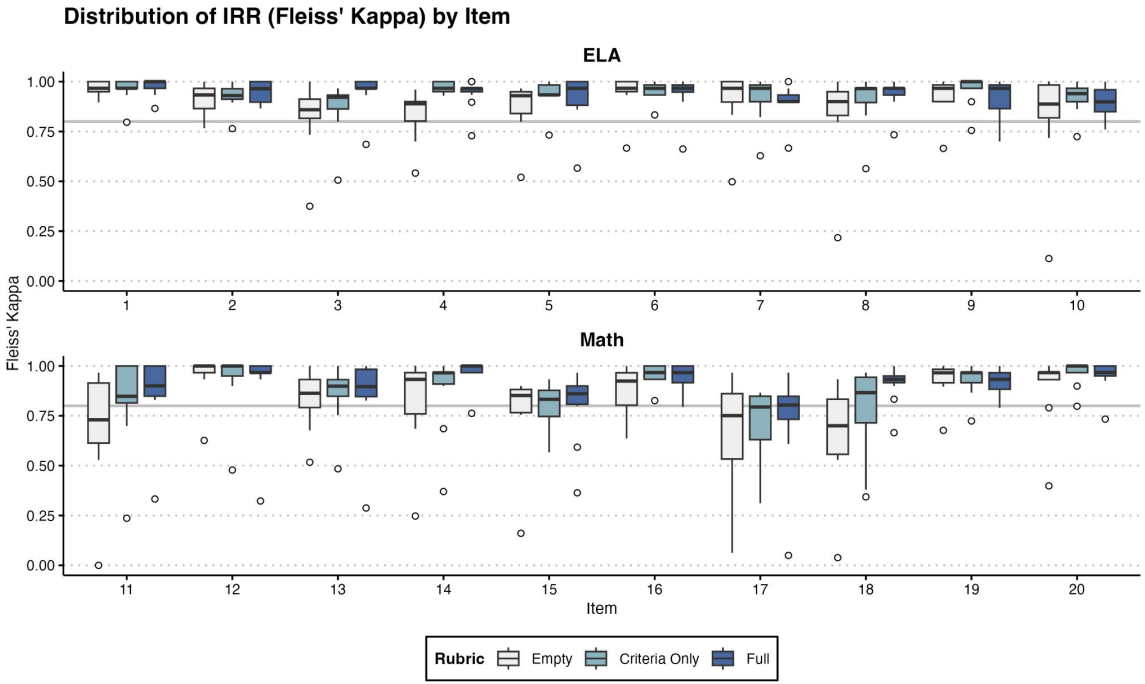
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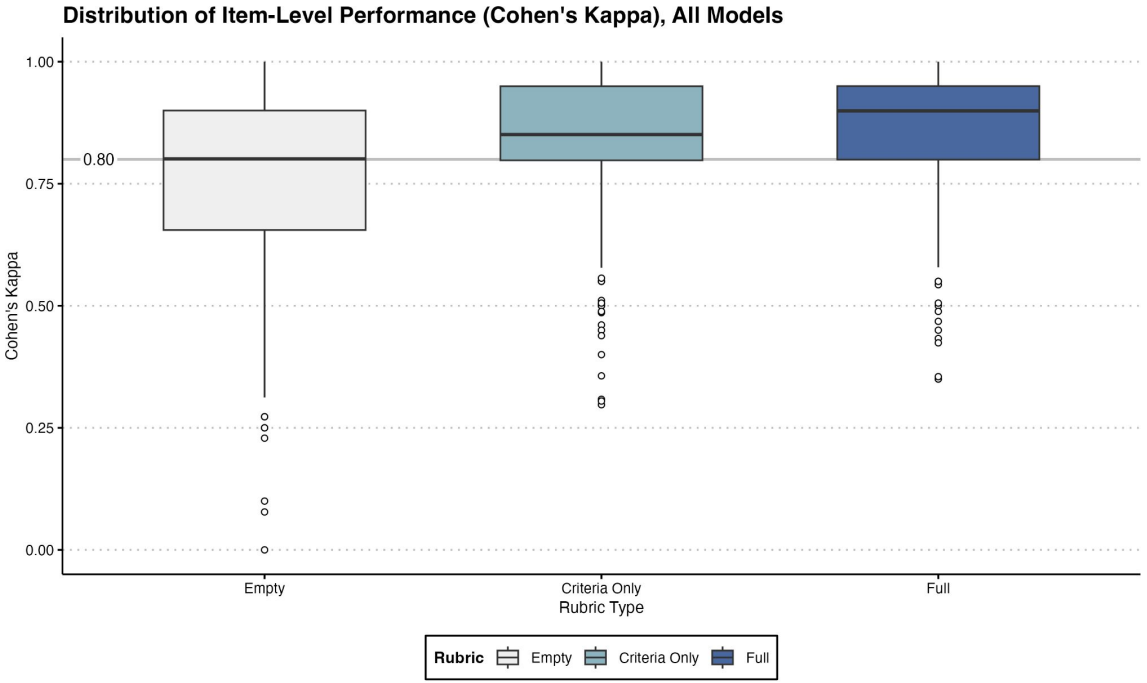
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# Performance by Rubric




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**Notes**

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**Results**

**ANOVA**: $F(2, 657) = 9.98$, $p < .001$; $\eta^2 = 0.03$

**Contrasts:**

***Criteria Only > Empty**

$M_{diff} = -0.072$, $SE = 0.016$, $t(657) = -4.54$, $p < .001$

***Full > Empty**

$M_{diff} = -0.089$, $SE = 0.016$, $t(657) = -5.64$, $p < .001$

Full v. Criteria Only

$M_{diff} = -0.017$, $SE = 0.016$, $t(657) = -1.10$, $p = .271$

# Overall Performance by Rubric and Model

| | | Perf. (Cohen's Kappa) | | |
|---|---|---|---|---|
| **Size** | **LM** | **Empty** | **Crit** | **Full** |
| Small | Claude-3-5 Haiku | .725 | .830 | .868 |
| | Gemini-1.5 Flash | .830 | .855 | .880 |
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| | GPT 4o-mini | .755 | .830 | .812 |
| | Llama 3.1 70b | .753 | .827 | .835 |
| | Llama 3.1 8b | .468 | .637 | .706 |
| Large | Claude 3-5 Sonnet | .825 | .837 | .867 |
| | Gemini 1.5 Pro | .845 | .872 | .870 |
| | GPT-4o | .808 | .880 | .910 |
| | Llama 3.1 405b | .756 | .852 | .877 |
| Frontier | Open AI o1 | .837 | .895 | .910 |

RQ2: How well do various LLMs score Short Answer Questions?

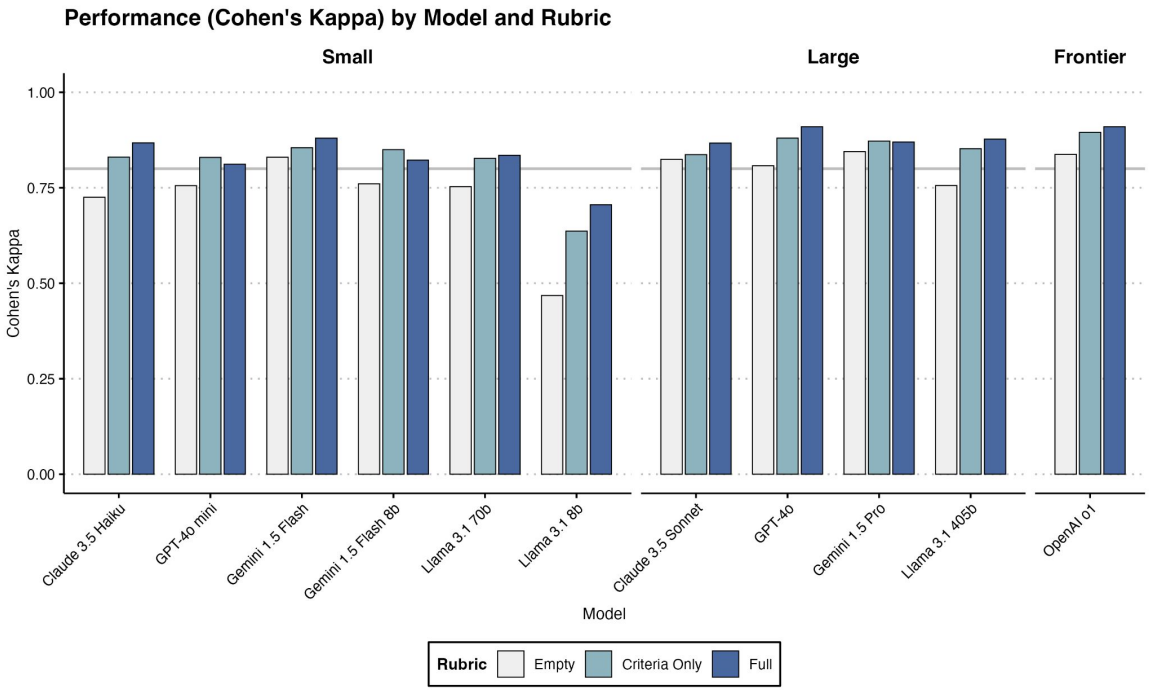
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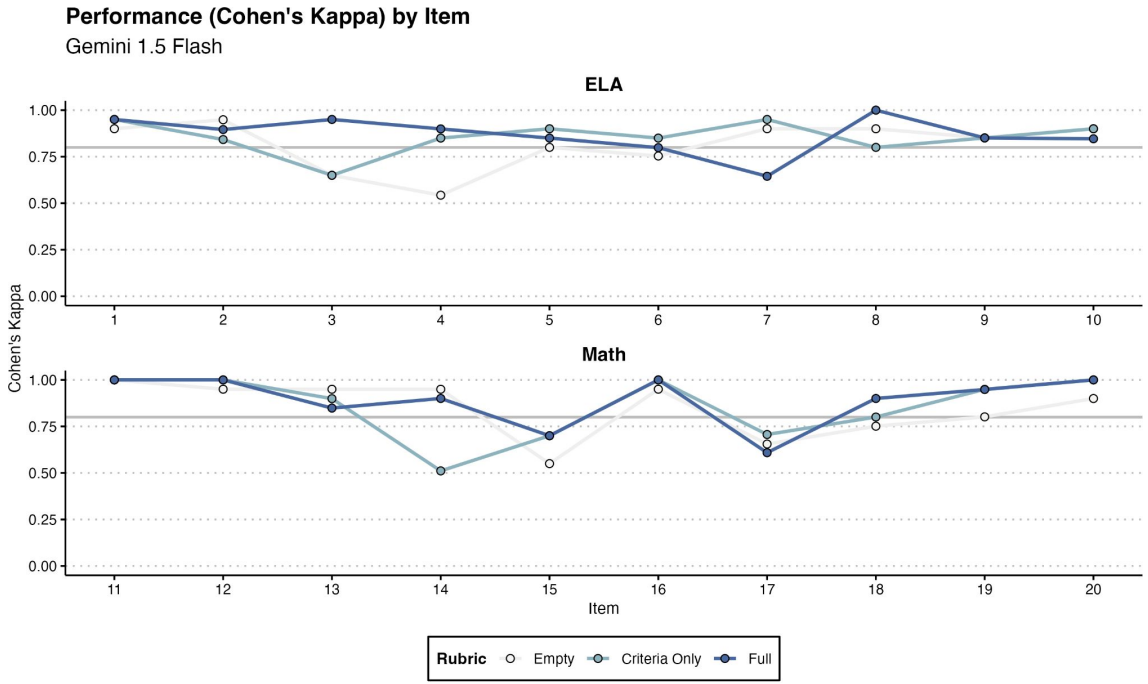
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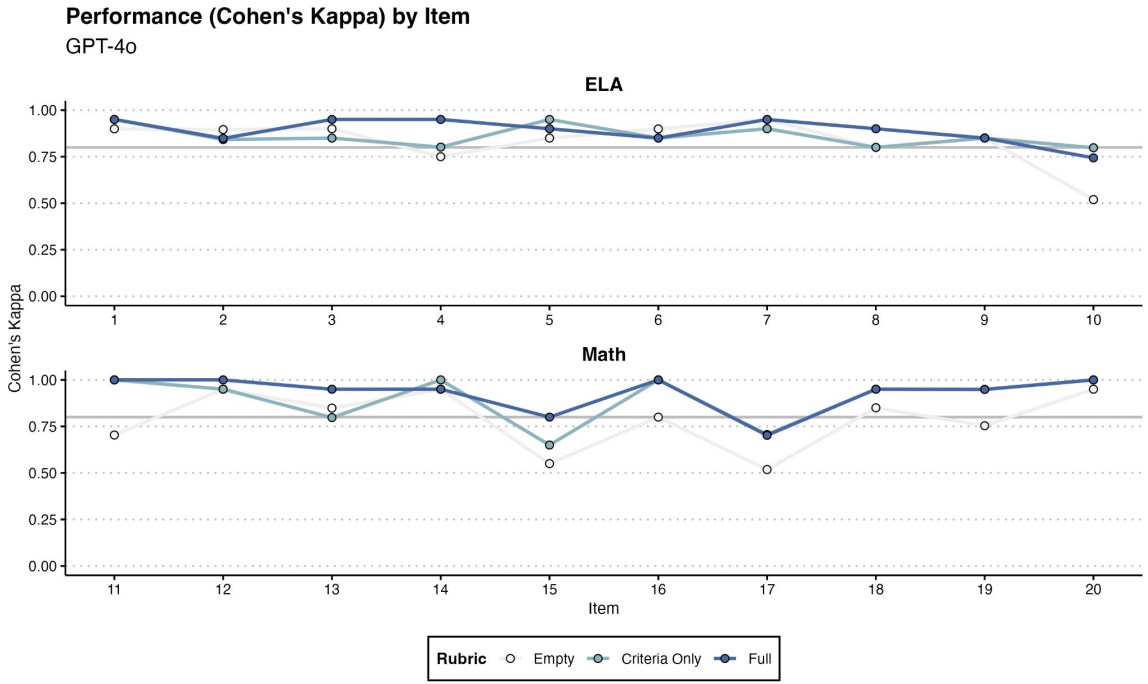
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**Observations**

- Performance varied by item
- Most items improved w/ rubrics
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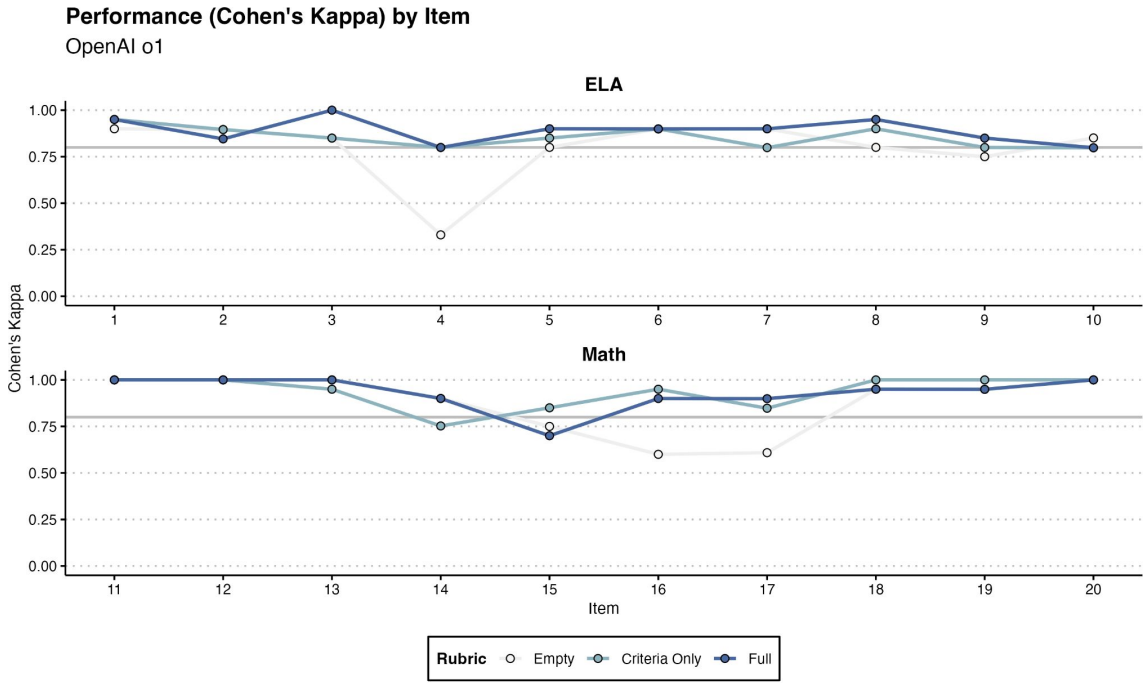
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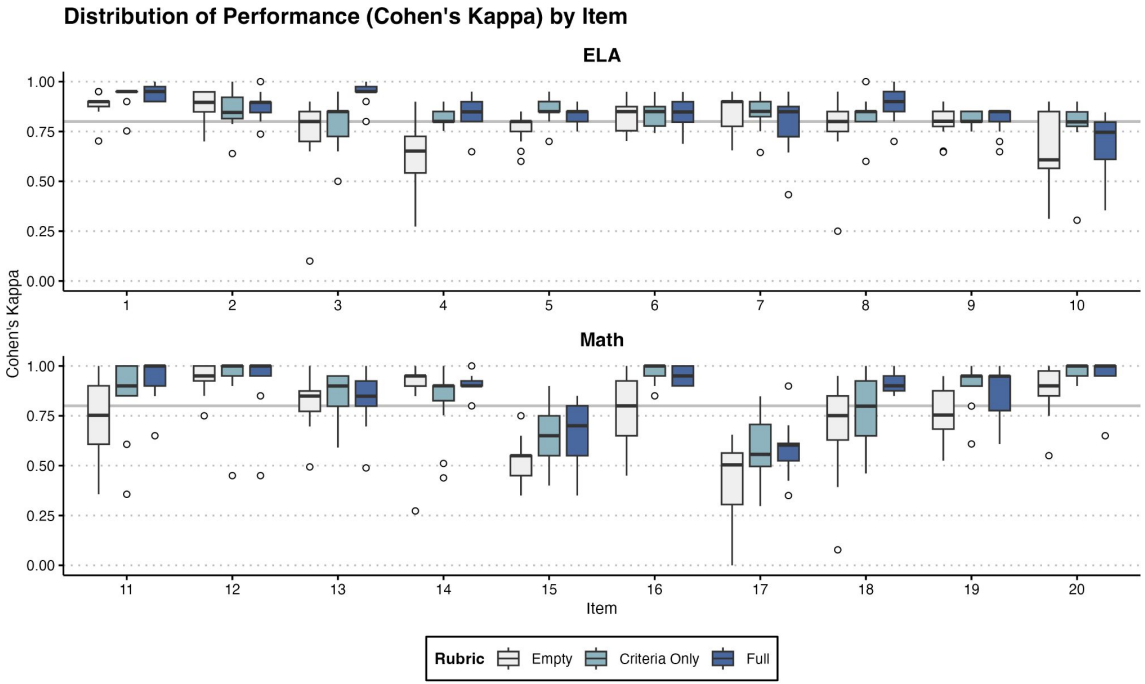
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# LLM Performance by Item Characteristics

Welch's Two Sample t-tests and ANOVA for group difference in Cohen's Kappa; Full Rubrics Only

**Domain**
Math | ELA

No effect of Domain on LLM performance.

$M_{diff}$ = -0.008, t(193.58) = -0.43, p = 0.666

**Response Length**
Shorter | Longer

LLMs tended to score items requiring shorter responses (<1 sentence) more accurately than items requiring longer responses (1+ sentences).

$M_{diff}$ = -0.08, t(217.98) = -4.75, p < .001,
Cohen's d = -0.64, 95% CI [-0.91, -0.37]

$M_{Shorter}$ = 0.90 > $M_{Longer}$ = 0.82

**Cognitive Demand**
Low | High

No effect of Cognitive Demand on LLM performance.

$M_{diff}$ = -0.02, t(217.21) = -1.00, p = 0.321

**Scoring Subjectivity**
Low | High

No effect of Scoring Subjectivity on LLM performance.

$M_{diff}$ = -0.02, t(180.72) = -0.79, p = 0.430

**Answer Scope**
Narrow | Moderate | Broad

LLMs tended to score items with narrow answer scopes more accurately than items with broad answer scopes.

**ANOVA**: F(2, 217) = 3.74, p = 0.025; $\eta^2$ = 0.03

**Contrasts:**
Narrow > Moderate
$M_{diff}$ = -0.034, SE = 0.026, t(217) = -1.308, p = .192

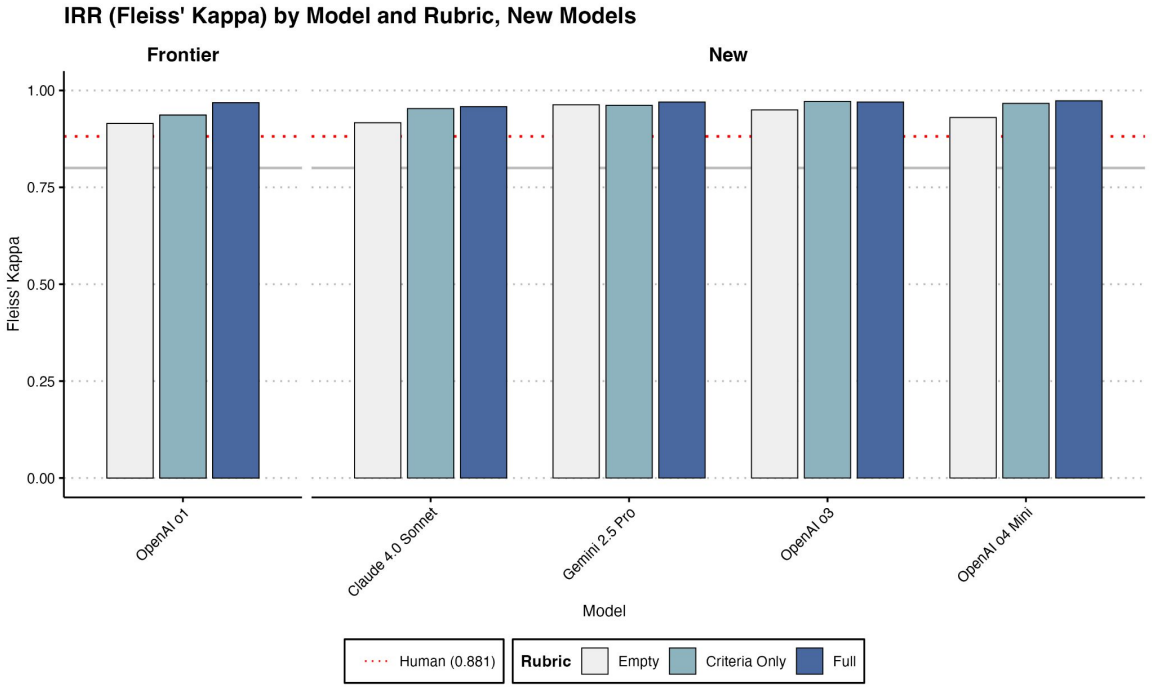
Moderate > Broad
$M_{diff}$ = -0.035, SE = 0.021, t(217) = -1.557, p = .099

***Narrow > Broad**
$M_{diff}$ = -0.069, SE = 0.026, t(217) = -2.661, p = .008

RQ3: What are the features of Short Answer Questions with the best and worst LLM performance?

# Bonus: More Models!!

# Overall Interrater Reliability by Rubric and (New) Model




## New LLMs

Claude 4.0 Sonnet
Gemini 2.5 Pro
OpenAI o3
OpenAI o4-Mini

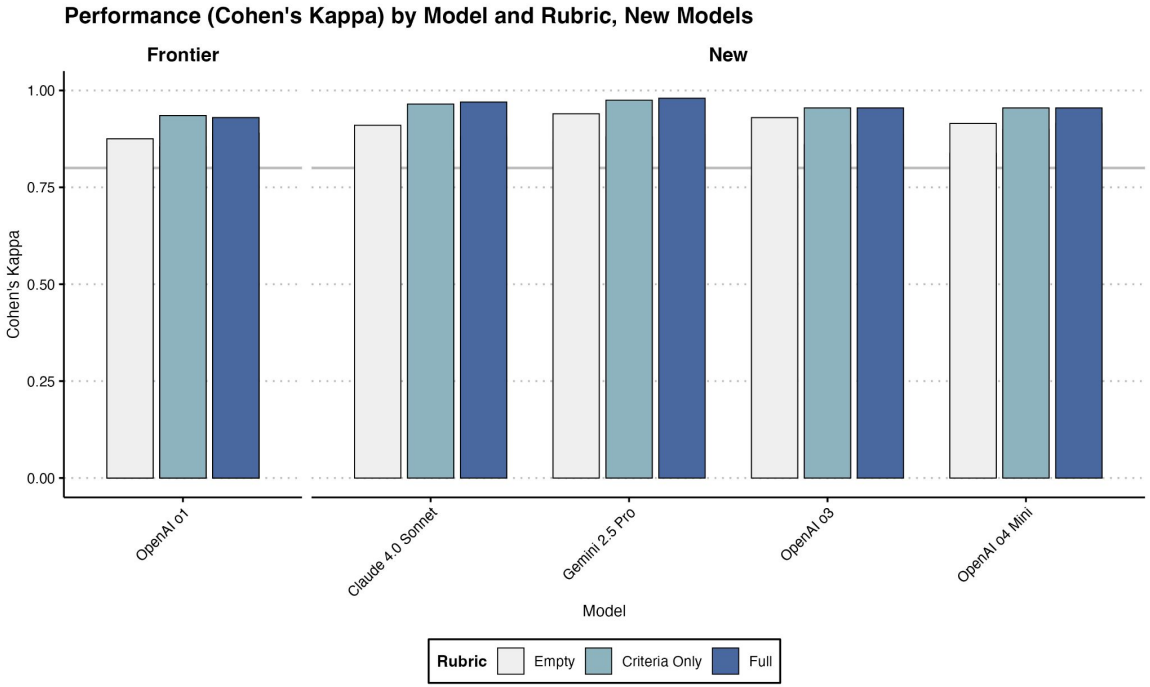
**Notes**

- Kappa > 0.80 represents Almost Perfect Agreement [15]
- Human IRR (Full Rubric) = 0.881

**Observations**

- New models have excellent IRR, on par with o1

# Overall Performance by Rubric and (New) Model




## New LLMs

Claude 4.0 Sonnet
Gemini 2.5 Pro
OpenAI o3
OpenAI o4-Mini

**Notes**

- Kappa > 0.80 represents Almost Perfect Agreement [15]

**Observations**

- New models have excellent performance, on par or better than o1

# Discussion

# Takeaways



1. **Off-the-shelf LLMs can score Short Answer Questions on-par with human scorers.**

2. **Including scoring criteria can improved model interrater reliability and performance against human ratings**
   *…but for some items and for some models, performance was very good even without scoring criteria or examples.*

3. **Including examples may not always improve interrater reliability and performance**
   *…but this may depend on the item.*

4. **Larger models tend to outperform smaller models**
   *…but some small models (e.g., Gemini 1.5 Flash) still performed very well.*

5. **LLMs tend to score items requiring shorter answers and narrow scope more accurately.**

6. **Using LLMs to score SAQs in development could help identify inconsistencies in scoring, leading to better rubrics.**

# Limitations

We used only synthetic response data.

Relatively small sample size.

Two scoring levels

# Future Directions

Collect more, authentic response data.

Compare actual response data against synthetic.

Evaluate performance of newer LLMs.

Experiment with different configurations of inputs (scoring criteria, model settings, number of judges, number of rating sets).



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# Thank you!

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github.com/Khan/llm-saq-scoring

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